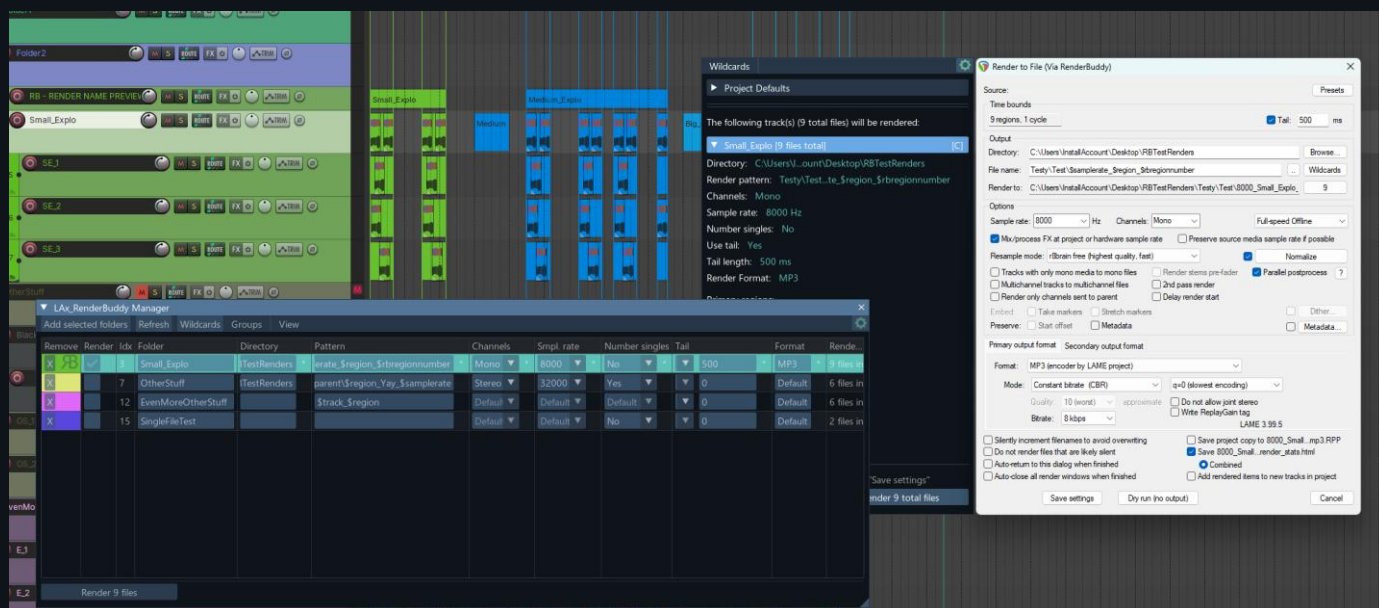




RenderBuddy

A lightning-fast, context-sensitive render solution for SFX and library creation in Reaper



Content

Content	2
Summary & Key features	1
Installation & requirements	1
Core concepts (If you read anything, let it be this!)	2
Automatic sound variation detection	2
Region/track-based rendering hybrid	2
Per-Folder and project default settings	2
QuickStart & basic functionality	3
First startup	3
Starting the render	3
Context-sensitive render via main action (LAX_RenderBuddy.lua)	3
Using the Manager to manage and render multiple folders	4
Main render dialog	4
Project Defaults	4
Per-Folder settings	4
Render Confirmation Window	5
Post-render window	6
Context-sensitivity	6
Settings (LAX_RenderBuddy - Settings.lua)	7
Advanced features	9
NameSwitches	9
Manager (LAX_RenderBuddy - Manager.lua)	10
Interface	10
Track blacklisting	12
Render selection grouping	12
Editing project default settings	13
Config Items	13
Custom Wildcards	13
Wildcard Manager (LAX_RenderBuddy – Wildcard Manager.lua)	14
Using Custom Wildcards	14
Baking regions	15
Using advanced Render Statistics	15

Tips and tricks	15
Reporting issues / giving feedback	16
Credits	16

Summary & Key features

RenderBuddy is a lightning-fast, context-sensitive render solution for SFX and library creation in Reaper 6 or later.

Key features include:

- **Automatic detection and clustering** of SFX variations.
- **Per-folder settings:** Render any number of folders in one go, with the ability to define different naming, target directories, file formats, sample rates and more for each individually
- **Context-sensitivity:** A single click can allow you to render based on selected items, selected tracks or track folders and their children.
- **No dedicated project setup required:** Regions are automatically created and named during the rendering process and RenderBuddy leaves no trace afterwards.
- **Custom Wildcards:** Define your own global or per-project wildcards and use them for file naming or even target directories!
- **Manager feature:** Persistently manage all of the above in a convenient interface!

Installation & requirements

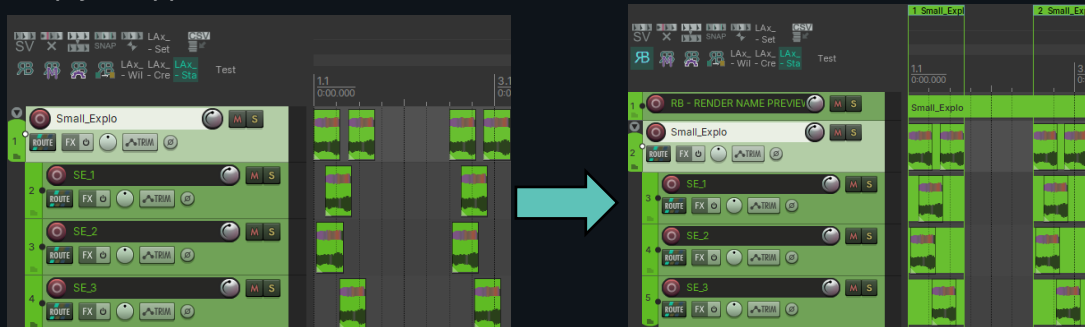
1. Install [ReaPack](#) (it's free!)
2. Click on *Extensions > ReaPack > Import repositories...*
3. Paste the repository URL and click *Ok*
 - a. Repository URL:
https://raw.githubusercontent.com/LAxemann/LAxemann_PremiumReaperScripts/refs/heads/main/index.xml
4. RenderBuddy should automatically guide you to downloading any further requirements upon launching any action.

Core concepts (If you read anything, let it be this!)

RenderBuddy is built to be intuitive, but some of its core concepts will help you to quickly find your way around and easily navigate its features.

Automatic sound variation detection

At its core, RenderBuddy automatically detects sound variations (stacks of items with gaps not longer than specified), creates regions and renders them. This happens in one go, so there's no need to set up any regions manually beforehand. After the render, all regions used for rendering simply disappear.



RenderBuddy can render multiple folders at once. When doing so, it will automatically render each folder selected for rendering in succession, creating and removing regions on the fly as it does so.

Region/track-based rendering hybrid

RenderBuddy could be described as a hybrid of region- and track-based rendering. The created regions are, by default, automatically named after the first track of its current render cycle.

Example: Your folder is called “Small_Explosion”. The created regions would get the same name, meaning the \$region wildcard would return “Small_Explosion”. Because each cycle is linked to a track, wildcards that are usually unavailable in region-based rendering are available in RenderBuddy, such as \$track, \$parent or even \$folders.

Per-Folder and project default settings

In addition to that, RenderBuddy allows you to set certain settings on a per-folder basis. Using either the [Manager](#) or [ConfigItems](#) you can, for example, persistently assign different output render directories and file names to individual folders and still render them in one go.

RenderBuddy supports the following settings on a per-folder basis:

- Directory
- File Name
- Channels
- Sample Rate
- Numbering of single files (RenderBuddy-own setting)
- Tail usage & length
- File format

If a setting is not set on a folder, it will use the RenderBuddy project default settings (Project Defaults) instead. These can be set either via the Manager or in the render settings window directly.

All of the aforementioned settings are not shared with Reaper’s default rendering. Changing e.g. the file name or directory field in either RenderBuddy or the “vanilla Reaper render” will not affect the other.

QuickStart & basic functionality

First startup

On first use of any RenderBuddy action, you will be asked to

- automatically run RenderBuddy’s startup action when launching Reaper (recommended) and
- automatically register all of RenderBuddy’s actions in a toolbar of your choice.

Both can also be done later through the settings (LAX_RenderBuddy – Settings.Lua).

Starting the render

There are two ways to go about rendering:

Context-sensitive render via main action (LAX_RenderBuddy.lua)

LAX_RenderBuddy.lua is a convenient choice for small projects or when quickly rendering individual elements of your session. The action is context-sensitive. More on that in the chapter: [Context-sensitivity](#). For a quick test, you could e.g. select a bunch of items or, with no items selected, select two individual folders.

Using the Manager to manage and render multiple folders

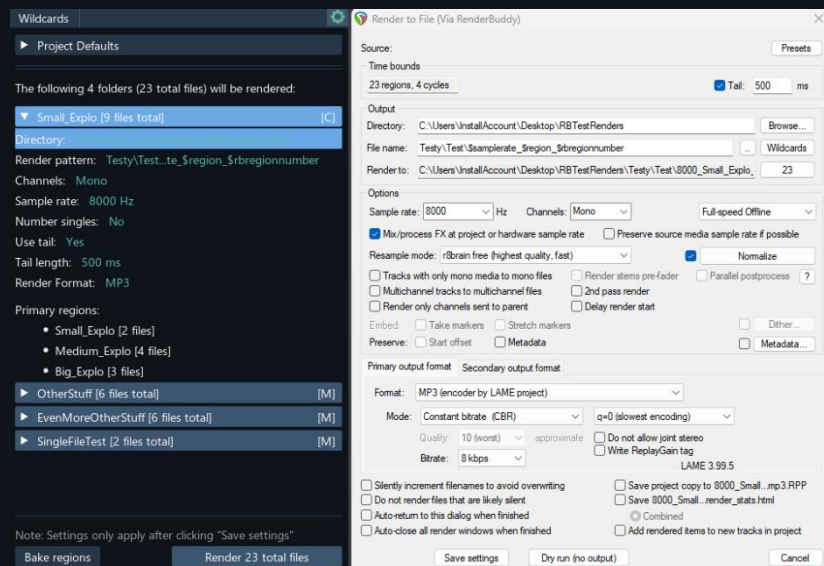
The Manager allows you to register folders and persistently save custom render settings on a per-folder basis. (More on that in the [Manager Section](#))

Simply run the LAX_RenderBuddy – Manager.lua action to open the manager, select one or multiple folders, and click “Add selected folders”. The folders will now show up in the Manager.

To render one or multiple folders, check the “Render” checkbox and click “Render X files” at the bottom left.

Main render dialog

Both the main action and the Manager will open up the main render dialog. The right-hand side shows the good ol’ Reaper render settings while the left-hand side shows everything related to RenderBuddy. Additionally, RenderBuddy should now have created a preview of the Clusters it detected.



Project Defaults

Starting with the left window, you’ll see at least two tabs: “Project Defaults” at the very top as well as one tab for each folder you’re rendering.

Clicking on “Project Defaults” allows you to set the settings RenderBuddy will use when a track does not have its own setting defined. To change any of the project defaults, simply change the value in Reaper’s main settings Window to the right and click on “Save settings”. The project default values should update immediately.

\$region and \$rbregionnumber wildcards

Note that, by default, the \$region wildcard will automatically use the name of the first track in the render selection. When doing a folder batch render, \$region will return the name of each folder.

RenderBuddy introduces the \$rbregionnumber wildcard. This will always properly number regions chronologically, even if you already have own regions created.

Per-Folder settings

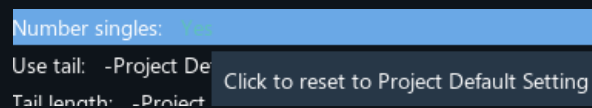
Clicking on any of the other tabs will switch RenderBuddy’s preview to the selected folder and also display its settings. If a setting is set to use the Project Defaults, it will simply show -Project

Default-. Hovering the mouse over this will show you the default value without having to switch to the Project Defaults tab.

If a folder is registered in the Manager, the tab will indicate this by displaying an [M] to the right. Any settings that differ from the Project Default or are explicitly set in the Manager will be highlighted in a different color.

To alter any of the settings, again, simply change them in Reaper’s render settings and click “Save settings”. If the folder is registered in the Manager, this will only change the settings for this folder. If the folder is not registered in the Manager (or using a Config Item, more on that later), the settings will be saved into the Project Defaults.

If you’d like to reset any setting to the project default, simply click on it:

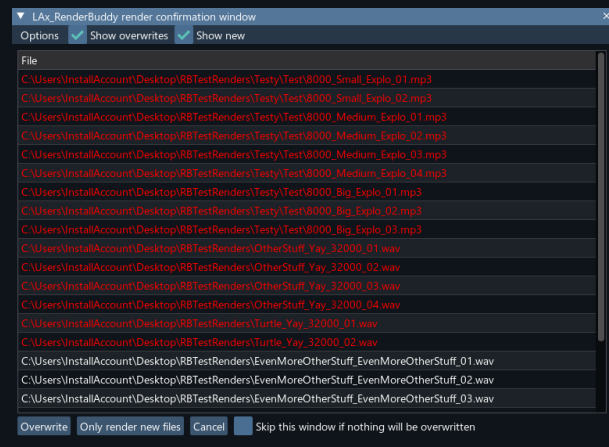


If your settings are to your liking, click “Render X total files” to get to the render confirmation window.

Render Confirmation Window

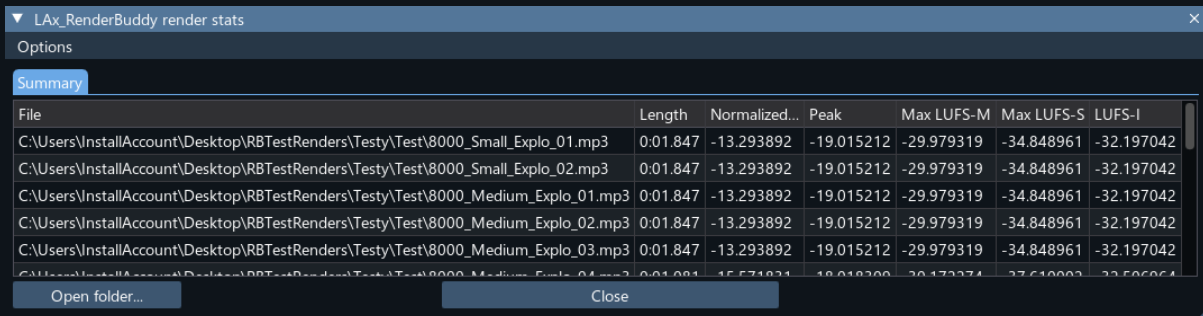
RenderBuddy features a custom confirmation window that shows the complete path of every created file (important given that each folder can now render into a different directory). Additionally, it will indicate already existing files by marking them red. If a render contains a mix of new and existing files, you can also choose to only render new files.

Confirming this window will start the actual render. If multiple folders were selected for rendering, RenderBuddy will sequentially render them one after another.



Post-render window

After the render, RenderBuddy will display a post-render window, listing all files together with render information. Since RenderBuddy allows you to render into multiple directories at once, clicking on “Open folder...” will give you a list of all folders that were rendered in.

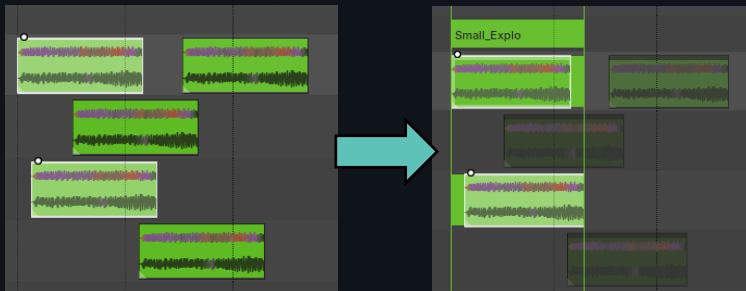


File	Length	Normalized...	Peak	Max LUFS-M	Max LUFS-S	LUFS-I
C:\Users\InstallAccount\Desktop\RBTestRenders\Testy\Test\8000_Small_Explo_01.mp3	0:01.847	-13.293892	-19.015212	-29.979319	-34.848961	-32.197042
C:\Users\InstallAccount\Desktop\RBTestRenders\Testy\Test\8000_Small_Explo_02.mp3	0:01.847	-13.293892	-19.015212	-29.979319	-34.848961	-32.197042
C:\Users\InstallAccount\Desktop\RBTestRenders\Testy\Test\8000_Medium_Explo_01.mp3	0:01.847	-13.293892	-19.015212	-29.979319	-34.848961	-32.197042
C:\Users\InstallAccount\Desktop\RBTestRenders\Testy\Test\8000_Medium_Explo_02.mp3	0:01.847	-13.293892	-19.015212	-29.979319	-34.848961	-32.197042
C:\Users\InstallAccount\Desktop\RBTestRenders\Testy\Test\8000_Medium_Explo_03.mp3	0:01.847	-13.293892	-19.015212	-29.979319	-34.848961	-32.197042
C:\Users\InstallAccount\Desktop\RBTestRenders\Testy\Test\8000_Medium_Explo_04.mp3	0:01.847	-13.293892	-19.015212	-29.979319	-34.848961	-32.197042

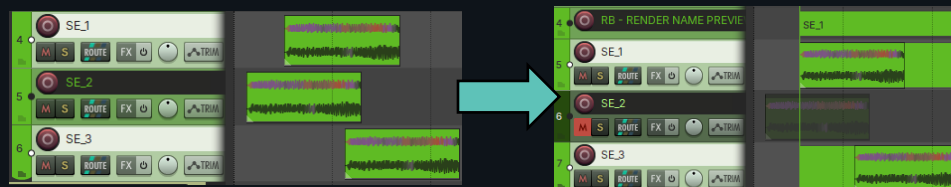
Context-sensitivity

When running the LAX_RenderBuddy.lua action, the rendered content depends on RenderBuddy’s context-sensitivity:

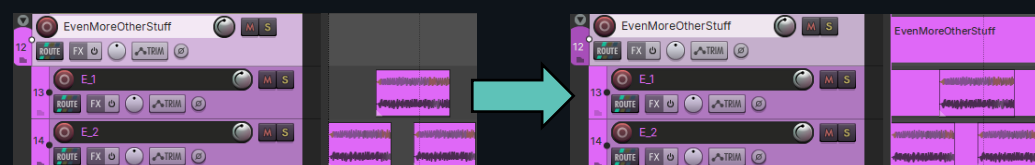
- If **NOTHING** is selected (no tracks and no items), it will create Clusters based on all items in the entire project.
- If **ITEMS** are selected, it will create Clusters based on the selected items.*



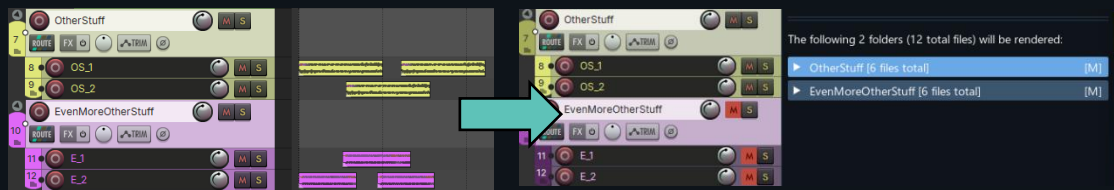
- If **ONE OR MULTIPLE TRACKS** and **NO ITEMS** are selected, it will create Clusters for all items on the selected tracks.



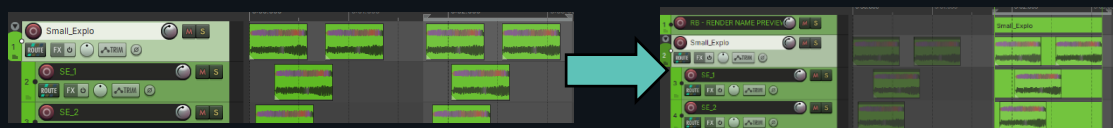
- If a **SINGLE FOLDER** is selected, it will create Clusters for all items within the folder.



- If **MULTIPLE FOLDERS** are selected, it will detect the Clusters and batch-render all selected folders in sequence.



- When doing track-based Cluster detection (no items selected), you can restrict the render area with Reaper's **TIME SELECTION**.**



* You can activate “Always render in track mode” in the settings if you want to never consider the current item selection.

** You can activate “Ignore time selection” in the settings if you want to never restrict the render boundaries to the time selection.

Settings (LAX_RenderBuddy - Settings.lua)

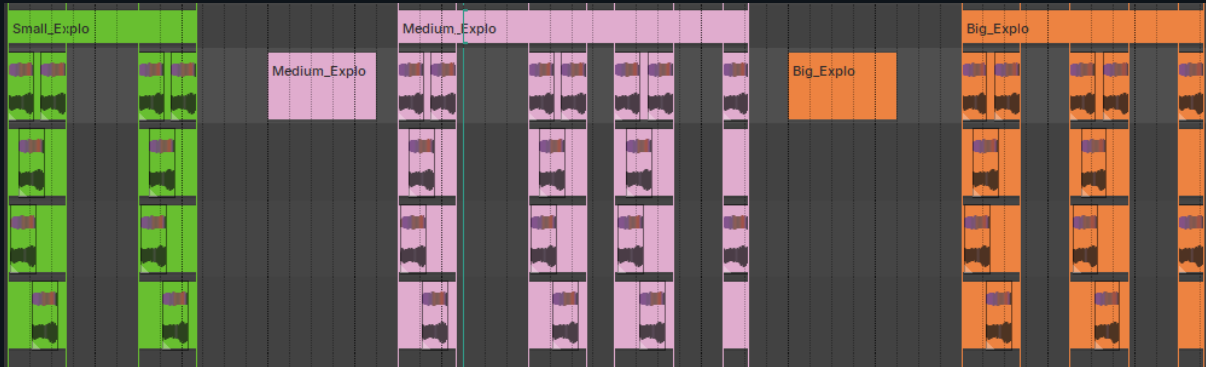
Setting	Description
Key Modifier: Use last settings	Using this key modifier will skip the render settings window and jump straight into the render confirmation.
Always render in track mode	If checked, RenderBuddy will always disregard the item selection and render in track mode.
Ignore time selection	If checked, RenderBuddy will never use the time selection as render region bounds.
Color-code items	If checked, items that are used for rendering will be color-coded
Collapse unused folders in preview	If checked, unused folders will get collapsed during preview.
Add visual helpers	If checked, visual helpers will be created to make the rendered areas more visible. Note: Creating these items is a costly task for Reaper and would stall in larger projects,

	which is why they automatically won't get created for (very, very) large selections.
Highlight/Animate elements	If checked, active elements like nameSwitches and folders will blink.
Set gap detection length to one beat	If checked, the gap detection link will be set to 1 beat no matter the bpm. Determines the maximum length of a gap between items before they are considered a separate sound and rendered into a new file.
Gap detection length	Determines the maximum length of a gap between items before they are considered a separate sound and rendered into a new file. Only available if "Set gap detection length to one beat" is turned off.
Preview track height	Determines the height of the preview track (0 = Disabled)
Default Render Pattern	Sets the default Render Pattern (= Filename pattern)
Number single variations	• If UNchecked, regions with only one file/variation won't get numbered.
Per-project Manager position	If checked, the Manager's position, size and docking status will be saved for each project. If UNchecked, the Manager window will keep size and position across all projects.
Flash tracks upon manager selection	If checked, tracks in the TCP will flash up when being selected in the Manager.
Check for updates on startup	If checked, running any RenderBuddy action will check for available updates once per session.
Automatically run Startup Action	If checked (RECOMMENDED), the startup action will automatically be launched when Reaper starts without the need to define your own startup action.
Add RenderBuddy actions to toolbar	Clicking this button will allow you to select a toolbar you'd like to register all RenderBuddy actions to. Note: Doing this on a completely untouched, default main toolbar will cause Reaper to only display RenderBuddy's actions. To circumvent this, just add any other action to the toolbar once and save.

Advanced features

NameSwitches

NameSwitches are items that allow you to switch the created region names on the same track. You could, for example, design multiple footstep surface variations on the same track. By placing NameSwitches and using the \$region wildcard, each surface variation can be rendered with a new name, and numbering will restart for each of them:



To create a NameSwitch, simply use the “LAX_RenderBuddy - Create NameSwitch.lua” action and enter a name in the following popup. This will create a NameSwitch item which you can drag around freely.

Once the render process hits the NameSwitch, numbering will reset to 1 and subsequent regions will be named after the NameSwitch.

There can be any number of NameSwitches per track, and each NameSwitch region will be shown in the Manager.

In order to edit/rename a NameSwitch, make sure the startup action is running (LAX_RenderBuddy – StartupAction.lua) and simply double-click the NameSwitch.

Manager (LAX_RenderBuddy - Manager.lua)

The Manager allows you to register folders and persistently save custom rendering settings on a per-folder basis.

Interface

Menu ribbon



Add Selected Folders: Registers the selected folders in the Manager

Refresh: Synchronizes the Manager with the project's current state

Wildcards: Opens the Wildcard Manager

Groups: Allows it to save and load render selection groups. (More in the dedicated chapter)

View: Allows switching between the main and blacklist window, changing how NameSwitch regions are displayed as well as toggling the project defaults bar on and off.

Cogwheel: Allows you to enable or disable table stretching and to access other settings.

Render Button: Starts the actual render. Requires at least one folder to have the "Render" checkbox ticked. Will then show the actual number of total files that will be rendered.

Per-folder settings columns



Note: Most of the following columns can be (un)hidden by right-clicking the column headers. If a Config Item is present on the track, each cell will indicate this with a * and setting changes will be applied to the Config Item, NOT the track itself.

Remove	Clicking the X will unregister the folder from the Manager. The cell will also have the color of the assigned track for easier identification. Additionally, if a Config Item is present, an icon will indicate this.
Render	Ticking the checkbox will add the folder to the render selection.
Idx	Index of the registered folder
Folder:	The name of the displayed folder. This field can also be used to quickly rename the actual folder within Reaper.
Directory	The render target directory for this track. Practically the same as Reaper's "Directory" field. When clicking the field, a ".." button will appear to the right, allowing you to select folder. An empty field will use the project's default directory
File name:	The pattern used to generate the file names. The same as Reaper's "File name" field. When clicking the field, a "\$" button will appear to the right, allowing you to pick both Reaper and RenderBuddy custom wildcards. An empty field will use the project's default pattern.
Channels	The number of channels the track will be rendered with. "Default" will use the project's default channel count.
Smpl. rate	The sample rate the track will be rendered in. "Default" will use the project's default sample rate.
Number singles:	Determines whether or not RenderBuddy automatically adds the number 1 to single files. Example if NOT on: • Sound_A • Sound_B_01 • Sound_B_02 Example if ON: • Sound_A_01 • Sound_B_01 • Sound_B_02
Tail	Allows activating a tail for rendering and settings its length. Same as Reaper's tail

	settings. Default will use the project's default tail settings.
Format	The rendering format, e.g. .MP3, .WAV etc. This can ONLY be changed in the render settings window.
Render region overview	Depending on the settings chosen in View -> Render Regions, the last columns will either show • Full: The individual render regions for each track (using NameSwitches can have multiple main regions per-track). • Overview: A written summary of the render regions • Don't show: Nothing will be shown

Track blacklisting

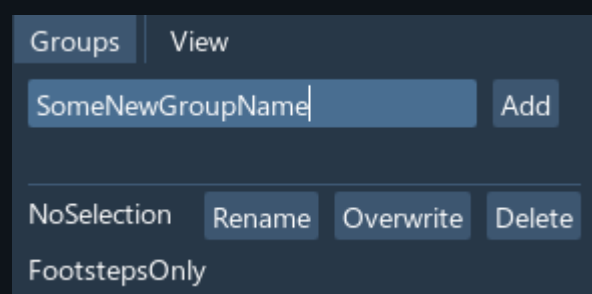
Individual tracks or whole folders can be blacklisted. This will fully exclude them from any rendering within RenderBuddy. This can be useful when you e.g. have a track with reference audio or video within your folder which you don't want to end up in the final render.

To do so, within the Manager, click View -> Active Tab and select "Blacklist".

You can then select one or multiple tracks or folders which you'd like to blacklist and click "Blacklist selected tracks".

Render selection grouping

Using the "Groups" button, you can create render selection Groups to save and load the manager's render selection state. This can be helpful when e.g. working on a large footstep project and you quickly want to only render folder that relate to "sand" surfaces without having to manually select all folders each time.



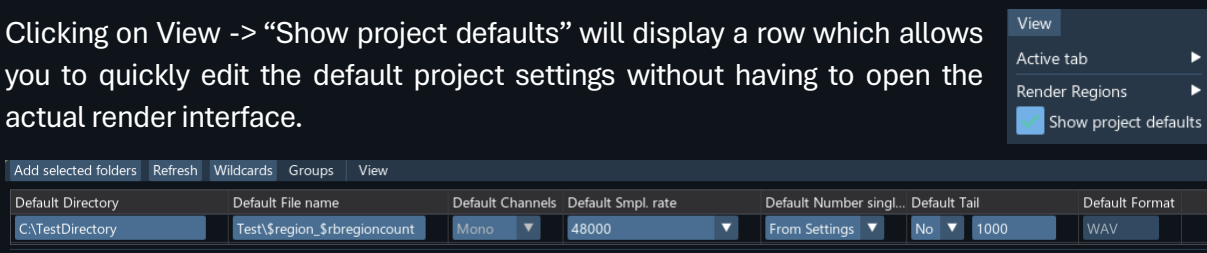
To **add a group**, simply select the folders you want to save, click "Group", enter a name into the text field and click "Add".

To **load a group**, simply click on the group's name in the list.

Hovering over a group in the list will also unveil the options to **rename**, **overwrite** or **delete** it.

Editing project default settings

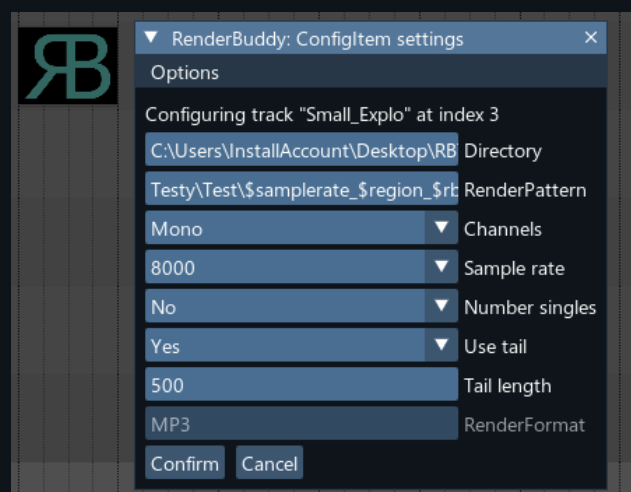
Clicking on View -> “Show project defaults” will display a row which allows you to quickly edit the default project settings without having to open the actual render interface.



Config Items

Config Items allow the definition of per-track render settings without the need to register them in the Manager. They are actual items within the timeline that can be freely moved to other tracks, copied etc.

To create a Config Item, select the track you want to create it on and use the “LAX_RenderBuddy - Create or edit ConfigItem.lua” action. A menu will appear which lets you adjust the Config Item’s render settings. For more info on those, please check the “Columns” section of the Manager chapter.



In order to edit/rename a Config Item, make sure the startup action is running (LAX_RenderBuddy – StartupAction.lua) and simply double-click the Config Item.

Alternatively, selecting a track that already has a Config Item and using the “LAX_RenderBuddy - Create or edit ConfigItem.lua” will also edit the existing one, rather than creating a new one.

If multiple Config Items are present on a track, the last (= rightmost) Config Item will be picked for editing.

Custom Wildcards

RenderBuddy features the possibility of defining your own Wildcards and then use them to name files or specify render directories. This can, for example, be incredibly useful when you have a large project in which multiple folders render into multiple different directories. Instead of copy/pasting the directory path into the directory field for each folder, you can simply define the path once as a wildcard and then use it multiple times. If the directory changes, you only need to edit the wildcard, not every single directory field.

Wildcard Manager (LAX_RenderBuddy – Wildcard Manager.lua)

New custom wildcards can be added via the Wildcard Manager.

To access it, simply use the the WildcardManager “LAX_RenderBuddy – Wildcard Manager.lua” action.

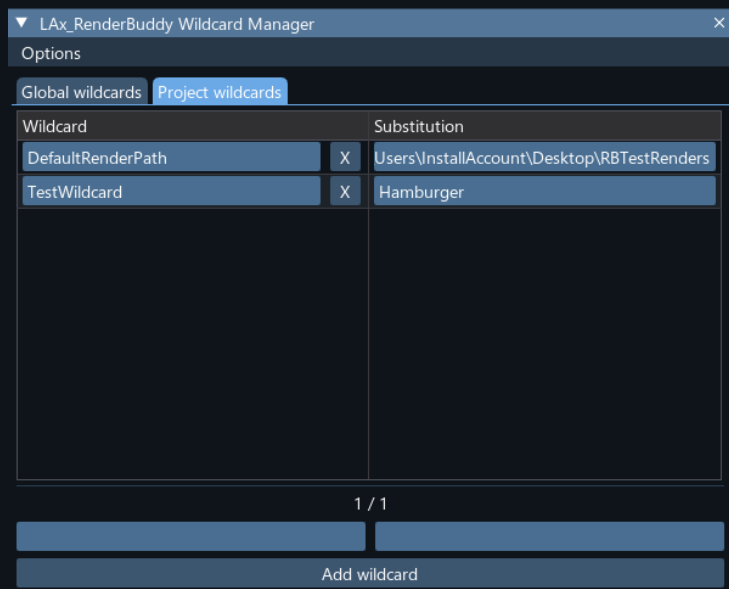
To save a wildcard, simply write the desired wildcard name into the bottom left input field and the desired substitution into the bottom right input field, then click “Add Wildcard”.

To edit a wildcard, simply change the wildcard or substitution name in the list.

To delete a wildcard, click the X button next to the wildcard’s name.

The Wildcard Manager has two dictionaries:

- “Global wildcards” holds wildcards that are accessible globally, on every project.
- “Project wildcards” holds wildcards that are only stored in the current project.



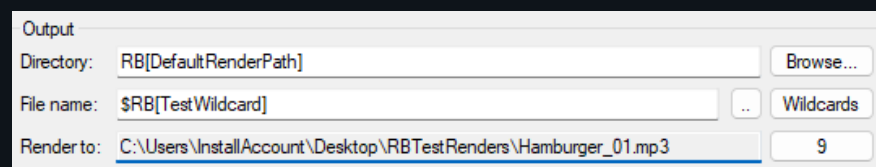
Additionally, you can change the number of displayed wildcards per page using the “Options” button.

Using Custom Wildcards

Custom wildcards can be used like regular wildcards, with the difference that they do not simply use the \$prefix. Instead, the format is

- \$RB[WILDCARDNAME] for the “File name” field
- RB[WILDCARDNAME] (without \$) for the “Directory” field

Example:



Baking regions

The main render window features a “Bake regions” button. This will permanently create the regions of the current render selection.

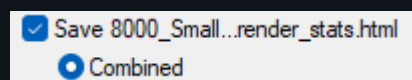
Note: Settings only apply after clicking “Save settings”

Bake regions

Render 6 total files

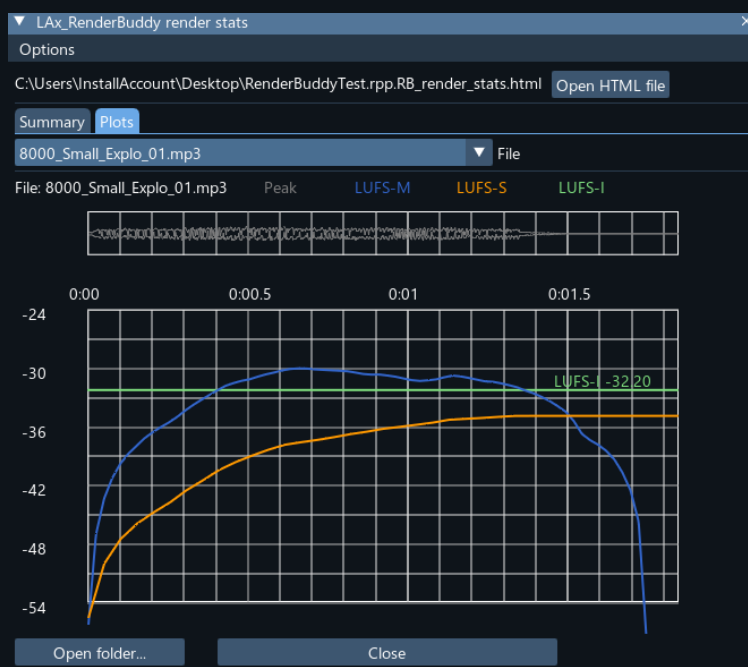
Using advanced Render Statistics

Advanced Render Statistics can be activated by clicking “Save XYZ...render_stats.html” and selecting “Combined” in Reaper’s render window:



Reaper will likely give you a warning, stating that you need to activate the saving of render statistics via the preferences. Click Yes.

After the render, the Post-render window will now have a “Plots” tab, allowing you to view a combined RenderStat output in RenderBuddy directly. The corresponding .html file will be saved in the project folder.



Tips and tricks

- You can hide individual Columns within the Manager by right-clicking the header row.
- The Manager supports CTRL as well as SHIFT selection. You can register or unregister multiple folders for rendering at once by first selecting them, then clicking the render checkbox of one of the folders.
- The Manager’s right-click context menu allows you to quickly render a single folder, register multiple folders for rendering at once, as well as unregistering multiple folders from the Manager at once.

Reporting issues / giving feedback

Issues and feedback can be reported/given here:

https://github.com/LAxemann/LAxemann_PremiumReaperScripts/issues

Credits

- Jan Köppen, Veronika Dubská, Jamie Lee, Manik Möllers, Andrej Novosad, Övünç Balcı, Alberto Sueri, Meghant Burman, XiaoXiao and Michał Mac for ideas and beta testing
- Cockos for the amazing DAW
- The amazing Reaper community, with special thanks to Christian Fillion for RealmGui and ReaPack in particular. Please consider donating to him when downloading ReaPack, Reaper would not be the same without him!